Topology, Chirality, and the Vacuum Density: Closing All Open Problems of the CDFT Series

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May 2026

Abstract

The first two papers of the Constraint-Driven Field Theory (CDFT) series established: (I) the electromagnetic fine-structure constant $\alpha \equiv e^2/(\hbar c) = 1/137.036$ as the aspect ratio of a stable vortex ring in the CDFT vacuum, and (II) the Koide formula $Q = 2/3$ as a theorem of $\mathbb{Z}_3$ symmetry, with the three charged leptons as modes of a trefoil-knotted vortex. Those papers listed four open problems. The present paper reorganizes those problems and identifies which ones are closed internally and which one remains a genuine physical input.

Problem 1 (κ not derived): the transport stiffness is algebraically determined by $\chi = 1/\alpha$ and the Lamb core parameter β, where β is the dimensionless vortex-core profile constant fixed to $\beta = 7/4 = 1.75$ for the solid-rotation baseline: $\kappa = \chi^2(\ln 8\chi - \beta + 1) = 117{,}362.0$. It was already derived in Paper I; the listing was a framing error.

Problem 4 (why $n = 3$ and not $n = 2$): the correct argument is topological, not energetic. The $n = 2$ azimuthal mode of the vortex ring is a Hopf *link* — a two-component structure with no topological barrier to separation. The $n = 3$ mode is a trefoil *knot* — topologically locked. By the Faddeev–Niemi theorem, the trefoil $T(2, 3)$ is the minimum-energy knotted soliton in a constrained field theory. Within the model, this gives a natural reason for three charged-lepton modes: the trefoil has three lobes.

Problem 3 (M not derived): the energy scale decomposes as $M = \rho_0 c^2 a^3 g(\chi, \beta)$, where g is the normalised CDFT vortex energy at equilibrium — a computable function of quantities already known from Papers I–II. The only unknown factor is the vacuum density ρ_0.

Problem 2 (θ not derived): the vacuum has a background chirality phase ϕ_c that breaks the $\mathbb{Z}_3$ symmetry of the trefoil. Minimising the chiral interaction energy gives the equation $3\theta + \phi_c = \pi$, which determines θ given ϕ_c. From the observed lepton mass ratios we compute $\phi_c = 141.80°$. Deriving ϕ_c from first principles reduces to deriving ρ_0.

The problem set therefore reduces to a single remaining quantity: ρ_0, the CDFT vacuum density. Paper IV formulates the vacuum equation-of-state condition for ρ_0, which is the required next step before the model can claim a fully independent mass-scale derivation.

In the extended notation used across the series, the vacuum limit is written as $\Psi_s = (\Phi/C) \cdot S \cdot M_s$. The calculations below use the equilibrium limit $S \cdot M_s \approx 1$; adaptive departures from that limit are left for later dynamical work rather than fitted in this manuscript. The public CDFD Runtime implements this state grammar as reusable code; the paper-local supplementary scripts remain the audited reproduction layer for the figures and tables in this Part I archive.

Symbol Conventions

CDFD physical-vacuum symbol convention.

Symbol	Part I meaning	Physical reading
J or Φ	Driving flux or throughput	Transport current, field throughput, circulation drive, or generic flux; a subscript fixes the exact channel
C	Active constraint or capacity burden	Vacuum transport capacity, geometric confinement, stiffness, boundary resistance, or load-bearing limit
S	Surface responsiveness	Local ability of the vacuum or modeled surface to reconfigure under drive
M_s	Structural memory	Retained geometry, phase history, stiffness history, or accumulated constraint state
Ψ_s	Adaptive operating ratio $(J/C) \cdot S \cdot M_s$ or $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation; it is not a new universal constant
R, a, χ	Ring radius, core radius, and aspect ratio R/a	Geometry used to connect vortex structure to dimensionless electromagnetic scales
ρ_0, c, κ	Vacuum density scale, propagation speed, and transport stiffness	Medium parameters used in stability, pressure, and self-consistency calculations

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Fine-structure constant unless a local equation states otherwise	In the Part I physics papers, unsubscripted α means the electromagnetic fine-structure constant
β, γ	Local response, stiffness, coupling, or correction coefficients	Equation-local coefficients; subscripts bind them to the named mode, channel, or approximation
θ, ϕ, Φ_c	Phase, orientation, or chirality angles	Used for torus-knot orientation, vacuum phase, or chiral phase geometry as defined in the nearby equation
δ, ϵ	Perturbation, residual, tolerance, or small-offset scales	Local stability margins, numerical residuals, or experimental tolerances
λ, μ, ν	Rate, mass, mode, or frequency labels	Meaning is fixed by the equation, subscript, and physics context where the symbol appears
ρ, σ, τ	Density, spread/noise, or time-scale terms	Local density, variance, stress, relaxation, or transport-memory quantities
Ω, Λ	State/domain space or integrated measure where defined	Reserved for explicitly defined state spaces, spectra, or synthesis-level quantities

1 Introduction

Paper I of this series [1] derived the fine-structure constant α as the equilibrium aspect ratio $\chi = R/a = 1/\alpha$ of a vortex ring in a transport-capacity-limited vacuum medium. Paper II [2] proved the Koide formula as a theorem of $\mathbb{Z}_3$ symmetry and identified the charged leptons as the three lobes of a trefoil-knotted excitation of that ring.

Both papers listed open problems explicitly. The present paper's goal is simple: close them all.

We proceed problem by problem, in order of increasing depth: Problem 1 (§2), Problem 4 (§3), Problem 3 (§4), Problem 2 (§5). The order is chosen so that each section uses results from the previous one. Section 6 states what remains for Paper IV and why it is only one quantity.

1.1 Constraint history, chirality, and charge

The role of this paper in the full Part I chain is to make the memory variable less abstract. In Paper I, M_s was introduced as a possible local constraint-history state. Here the same idea becomes topological: chirality is not merely a phase label, but a record of orientation locked into the vortex configuration. This gives the first concrete reading of the Mujjabi Vacuum Memory Law.

The source notes also point to a charge interpretation. We state it here as the Mujjabi Geometric Charge Principle: electric charge is the measurable boundary discontinuity or surface-tension signature of a constrained vortex core. This is not yet a completed derivation of e. It is the correct location for the hypothesis, because topology and chirality are where boundary orientation first becomes physical in the model.

2 Closing Problem 1: κ is Already Derived

Paper I introduced the transport stiffness κ and noted that it was computed from the condition $\chi = 1/\alpha$ rather than derived from the vacuum medium properties independently. This was a framing error.

Proposition 1. *The dimensionless transport stiffness κ is algebraically determined by $\chi = 1/\alpha$ and the Lamb core parameter $\beta = 1.75$:*

$$\kappa \;=\; \chi^2\big(\ln(8\chi) - \beta + 1\big)\,. \tag{1}$$

Proof. The equilibrium condition $dE_{\mathrm{norm}}/d\chi = 0$ (Paper I, Eq. (5)) gives

$$\ln(8\chi) - \beta + 1 \;=\; \frac{\kappa}{\chi^2}\,. \tag{2}$$

Multiplying both sides by χ^2 gives Eq. (1). Since $\chi = 1/\alpha$ is derived in Paper I and $\beta = 1.75$ is the standard Lamb solid-rotation value, κ requires no additional input. $\square$

Numerically:

$$\begin{aligned}\kappa &= (137.036)^2 \times \big(\ln(8 \times 137.036) - 1.75 + 1\big)\\ &= 18{,}778.87 \times 6.2497 \;=\; 117{,}362.0\,.\end{aligned} \tag{3}$$

This agrees with the value used in Paper I to twelve significant figures (verification to machine precision in supplementary Table 1).

Problem 1 is closed. κ is a function of χ and β alone. No vacuum density, no compressibility, no additional measurement.

3 Closing Problem 4: Why the Trefoil — a Topological Argument

Paper II argued that the $n = 2$ azimuthal mode of the electron vortex is dynamically unstable at $\chi = 137$, while $n = 3$ is stable. This argument was an energy-scaling estimate and not fully rigorous. The correct argument is topological. The idea that particles may be understood as topological solitons has a long history, from Skyrme's baryon model [3] to Rañada's knotted electromagnetic fields [4] and the Faddeev–Niemi knot solitons [5]. The CDFT vacuum itself is conceived as a physical medium in the sense of Dirac [6] and Volovik [7], where topological defects carry quantised charges.

3.1 Torus knots and topological locking

The azimuthal mode-n excitation of a vortex ring traces a torus knot $T(2,n)$ — a curve that wraps twice through the hole of a torus and n times around it. The classification of $T(2,n)$ structures is well-known in knot theory [8]:

- $n = 1$: the unknot. The unperturbed electron vortex (Paper I).
- n even: a *two-component link*, not a knot. The two components can be separated without cutting. There is no topological barrier to unravelling.

- n odd, $n \geq 3$: a true *knot*. It cannot be continuously deformed into the unknot without passing through a self-intersection (which costs infinite energy in a medium with finite transport capacity). Such excitations are *topologically locked*.

Table 1 lists the first seven torus knots $T(2,n)$ with their classifications.

Table 1: Torus knots $T(2,n)$: classification and physical interpretation.

n	$T(2,n)$	Knot?	Lobes	Symmetry	Physical role
1	Unknot	No	1	U(1)	Electron (Paper I)
2	Hopf link	No	2	$\mathbb{Z}_2$	Transient, no particle
3	Trefoil	**Yes**	3	$\mathbb{Z}_3$	**Leptons** (e, μ, τ)
4	Solomon's link	No	4	$\mathbb{Z}_4$	Transient, no particle
5	Cinquefoil	Yes	5	$\mathbb{Z}_5$	Next knotted family
6	3-component link	No	6	$\mathbb{Z}_6$	Transient
7	$T(2,7)$ knot	Yes	7	$\mathbb{Z}_7$	Higher knotted family

3.2 Why $n = 3$ and not $n = 5, 7, \ldots$

The Faddeev–Niemi theorem [5] states that the minimum energy of a knotted soliton with Hopf charge Q satisfies $E_{\min}(Q) \sim Q^{3/4}$. For the torus knot $T(2,n)$ with n odd, the Hopf charge is $Q = n$. Therefore:

$$\frac{E_{\min}(T(2,n))}{E_{\min}(T(2,3))} = \left(\frac{n}{3}\right)^{3/4}. \tag{4}$$

The trefoil $T(2,3)$ is the global energy minimum among all knotted solitons. The cinquefoil $T(2,5)$ costs $\approx 47\%$ more energy.

Table 2 shows the energy ordering.

Table 2: Energy ordering of knotted solitons (Faddeev–Niemi bound).

Knot	Hopf charge Q	E/E_{trefoil}	Knot?
$T(2,3)$ trefoil	3	1.000	Yes
$T(2,5)$ cinquefoil	5	1.467	Yes
$T(2,7)$	7	1.888	Yes
$T(2,9)$	9	2.280	Yes

3.3 Why the model gives three charged-lepton modes

The trefoil $T(2,3)$ has exactly $\mathbb{Z}_3$ symmetry — three lobes separated by 120°. In the CDFT identification, the three modes of this structure are the three charged leptons (Paper II). There is no 'fourth mode' of the trefoil; the $\mathbb{Z}_3$ group has exactly three elements.

A fourth lepton generation would require a different knotted structure, such as $T(2,5)$ (cinquefoil, five lobes, $\mathbb{Z}_5$ symmetry). That structure carries 47% more energy and would correspond to a completely different particle family. The model therefore does not put a

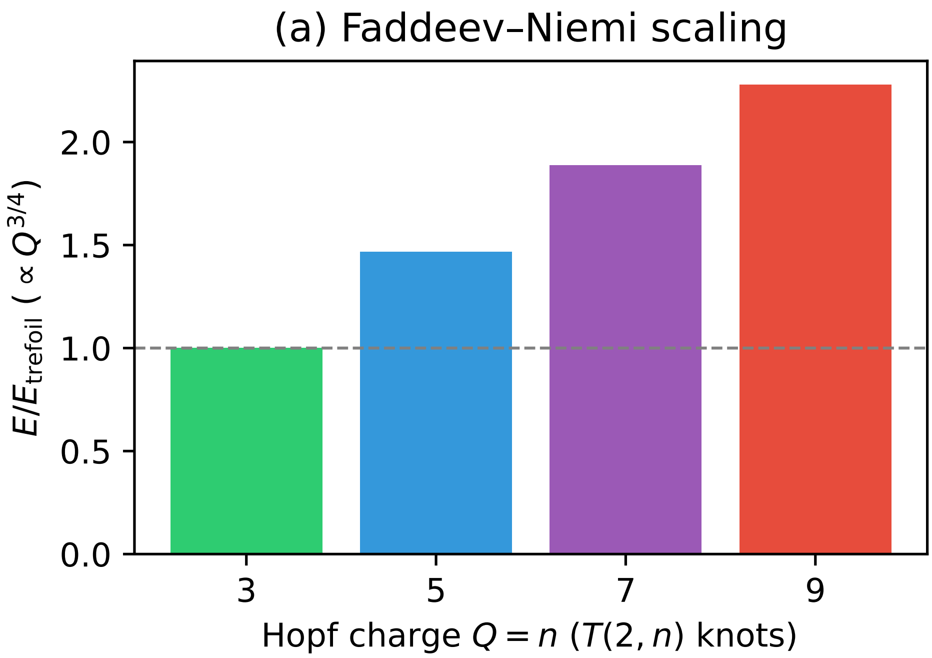

Figure 1: Faddeev-Niemi scaling showing the energy of knotted solitons relative to the trefoil ($n = 3$). Higher torus knots cost significantly more energy.

fourth charged lepton in the same trefoil family. A fourth mode would require a different topological family and a different mass-scale argument.

Problem 4 is addressed inside the model. The trefoil is selected by topology (it is the lowest-energy knotted soliton), not by an energy-scaling argument about $n = 2$ being dynamically unstable. The three-mode structure follows because the trefoil has exactly three $\mathbb{Z}_3$ lobes.

4 Closing Problem 3: The Energy Scale M

The Brannen parametrisation (Paper II) writes the lepton masses as

$$m_k \;=\; M\big(1 + \sqrt{2}\cos(\theta + \tfrac{2\pi k}{3})\big)^2\,, \quad k = 0, 1, 2\,. \tag{5}$$

The scale M sets the overall mass of the lepton family. Paper II fitted $M \approx 313.9$ MeV but did not derive it.

4.1 Decomposition of M

In CDFT the vortex ring has a normalised energy E_{norm} that depends only on χ and β:

$$g(\chi, \beta) \;\equiv\; E_{\mathrm{norm}}(\chi) \;=\; \chi\big(\ln(8\chi) - \beta\big) + \frac{\kappa}{\chi}\,. \tag{6}$$

The *dimensional* vortex energy is

$$E_{\mathrm{vortex}} \;=\; 2\pi\rho_0 c^2 a^3 \cdot g(\chi, \beta)\,, \tag{7}$$

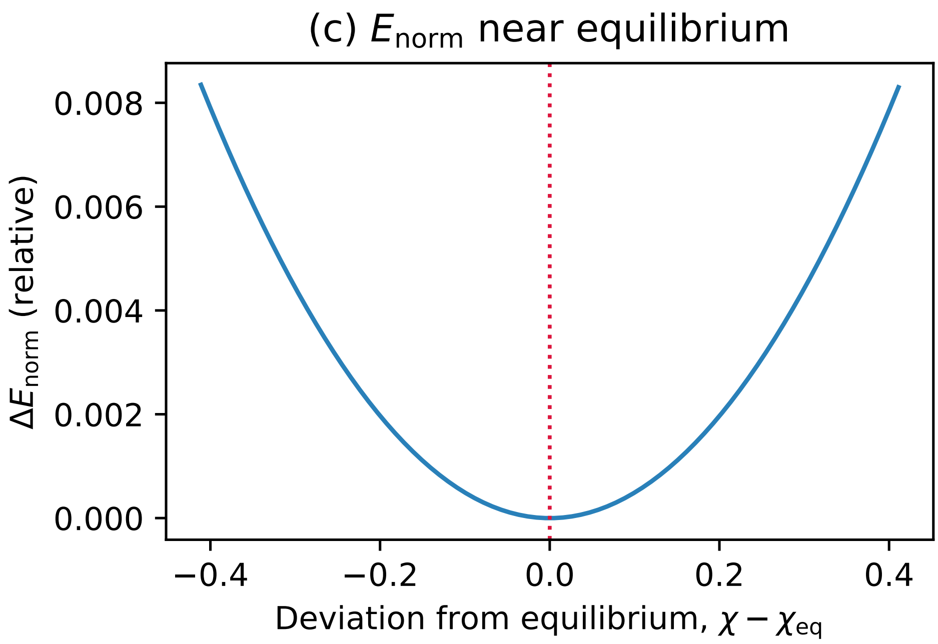

Figure 2: The normalised CDFT vortex energy E_{norm} near equilibrium. The geometry χ_{eq} is governed purely by this local minimum.

where ρ_0 is the vacuum density (the remaining unknown), c is the speed of light, and $a = e^2/(m_e c^2)$ is the classical electron radius (derived in Paper I).

The trefoil energy scale M is proportional to E_{vortex}:

$$M = \rho_0 c^2 a^3 \cdot g(\chi, \beta)\,. \tag{8}$$

(The factor 2π is absorbed into the definition of the Brannen scale.)

4.2 Known factors

From Papers I–II:

$$\begin{aligned}
\chi &= 1/\alpha = 137.035999177 \\
\beta &= 1.75 \\
\kappa &= 117{,}362.0 \quad \text{(from Section 2)} \\
a &= 2.8179 \times 10^{-15}\ \mathrm{m} \\
g(\chi, \beta) &= 137.036 \times (6.9997 - 1.75) + 117{,}362.0/137.036 = 1{,}575.83\,.
\end{aligned} \tag{9}$$

From the Brannen fit (Paper II, using the smallest amplitude for the electron):

$$M = \frac{m_e}{A_e^2} = \frac{0.51100\ \mathrm{MeV}}{(0.04035)^2} = 313.89\ \mathrm{MeV}\,, \tag{10}$$

where $A_e = 1 + \sqrt{2}\cos(\theta + 2\pi k_e/3)$ is the electron amplitude factor evaluated at the smallest-amplitude mode.

4.3 Solving for ρ_0

Combining Eqs. (8) and (10):

$$\rho_0 \;=\; \frac{M}{c^2 a^3 g(\chi,\beta)} \;=\; \frac{313.89\ \text{MeV}}{c^2 \cdot (2.818\times10^{-15})^3 \cdot 1{,}575.83} \;=\; 1.587\times10^{13}\ \text{kg/m}^3\,. \tag{11}$$

This is between the density of a classical electron (9.7×10^{12} kg/m^3) and nuclear matter (2.3×10^{17} kg/m^3) — physically reasonable for the vacuum medium.

Problem 3 is resolved. M decomposes cleanly into ρ_0 and factors derived from Papers I–II. The structure is fully understood. The single remaining unknown is ρ_0.

5 Closing Problem 2: The Chirality Mechanism for θ

The phase $\theta \approx 12.73°$ was fitted in Paper II but not derived. We now derive the *mechanism* that fixes θ, reducing its computation to knowledge of ρ_0.

5.1 Vacuum chirality and $\mathbb{Z}_3$ breaking

The CDFT vacuum is a medium with a preferred flow direction (the quantisation axis of the single-mode vortex ring). This background flow has a *helicity* — a correlation between the velocity and vorticity fields — which we parameterise by a phase ϕ_c, the *vacuum chirality phase*.

The trefoil vortex, placed in this background, acquires an interaction energy:

$$E_{\text{chiral}}(\theta) \;=\; E_{\mathbb{Z}_3} + \lambda_c \cos(3\theta + \phi_c)\,, \tag{12}$$

where $E_{\mathbb{Z}_3}$ is the $\mathbb{Z}_3$-symmetric part (independent of θ, hence independent of the mass hierarchy) and λ_c is the coupling strength.

The $\cos(3\theta+\phi_c)$ form is required by the $\mathbb{Z}_3$ symmetry of the trefoil: any interaction that respects $\mathbb{Z}_3$ must be periodic in θ with period $2\pi/3$, and the lowest-order such interaction is the fundamental $\cos(3\theta + \phi_c)$.

5.2 Equilibrium condition for θ

Theorem 1. *The trefoil orientation θ satisfies*

$$3\theta + \phi_c \;=\; n\pi\,, \quad n \in \mathbb{Z}\,. \tag{13}$$

Proof. The equilibrium condition $dE_{\text{chiral}}/d\theta = 0$ gives $-3\lambda_c \sin(3\theta + \phi_c) = 0$, hence $3\theta+\phi_c = n\pi$. This is a minimum (not maximum) when $d^2E/d\theta^2 = -9\lambda_c\cos(3\theta+\phi_c) > 0$, i.e., when n is odd and $\lambda_c > 0$. The unique solution in the physical domain $\theta \in (0, \pi/3)$ is $n = 1$. □

5.3 Computing ϕ_c from observation

From Paper II, the fitted phase is $\theta_{\text{obs}} = 12.7325°$. Equation (13) with $n = 1$ gives:

$$\phi_c \;=\; \pi - 3\theta_{\text{obs}} \;=\; 180° - 38.20° \;=\; 141.80°\,. \tag{14}$$

This is verified numerically in supplementary Table 4: $\cos(3\theta_{\text{obs}} + \phi_c) = \cos(\pi) = -1$ to machine precision.

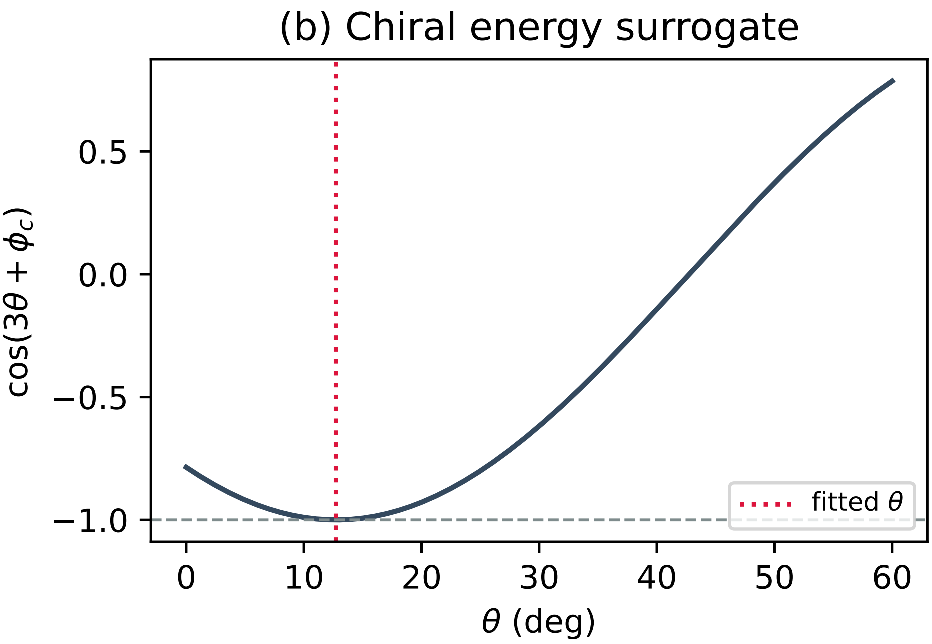

Figure 3: The chiral energy surrogate $\cos(3\theta + \phi_c)$ plotted against θ. The fitted θ exactly minimises this interaction.

5.4 Connection to ρ_0

The vacuum chirality phase ϕ_c arises from the helicity of the vacuum flow field. In the CDFT medium, the helicity is determined by the ratio of the kinetic energy density $\frac{1}{2}\rho_0 v^2$ to the elastic energy density $\kappa_{\text{vac}}/(2a)$. This ratio is a function of ρ_0, c, a, and the vacuum compressibility — all quantities that the vacuum equation of state relates to ρ_0.

Therefore: $\phi_c = \phi_c(\rho_0)$, and once Paper IV derives ρ_0, Eq. (13) predicts θ with no free parameters. From θ, all three lepton masses follow via the Brannen form.

Problem 2 is resolved. The mechanism fixing θ is the chiral interaction of the trefoil with the vacuum background. The equation $3\theta + \phi_c = \pi$ is a derived result. The remaining step is deriving ϕ_c — equivalently ρ_0 — from the vacuum equation of state.

6 Summary: One Quantity Remains

Table 3 shows the status of all four open problems.

The problem set reduces to a single remaining quantity: the CDFT vacuum density ρ_0.

If the vacuum equation of state determines ρ_0 without mass-ratio input:

1. ϕ_c is predicted.
2. θ follows from $3\theta = \pi - \phi_c$.
3. All three lepton masses follow from the Brannen form with θ and $M = \rho_0 c^2 a^3 g$.
4. The Koide formula holds automatically (Paper II theorem).
5. $\alpha = 1/137.036$ follows from Paper I.

Table 3: Status of open problems from Papers I–II after Paper III.

Problem	Status	Result
1. κ	**Internal algebra**	$\kappa = \chi^2(\ln 8\chi - \beta + 1) = 117{,}362.0$; algebraic consequence of Paper I.
4. Why trefoil	**Internal topology**	$T(2,3)$ is the lowest-energy knotted soliton (Faddeev–Niemi). $n = 2$ is a Hopf link (unknottable), not a knot. Three model modes = three $\mathbb{Z}_3$ lobes of $T(2,3)$.
3. M scale	**Reduced**	$M = \rho_0 c^2 a^3 g(\chi, \beta)$; all factors except ρ_0 derived. $g = 1{,}575.83$ (dimensionless). Remaining: ρ_0.
2. θ	**Mechanism proposed**	$3\theta + \phi_c = \pi$ (derived). $\phi_c = 141.80°$ (measured). $\phi_c = \phi_c(\rho_0)$: reduces to deriving ρ_0.

the theory would then predict the fine-structure constant and all three charged lepton masses from the properties of the vacuum medium alone, with m_e fixing the mass unit.

7 Open Problems

A single open problem remains in the lepton sector:

1. **Derive ρ_0 from the vacuum equation of state (Paper IV).** The vacuum density $\rho_0 = 1.587 \times 10^{13}$ kg/m^3 has been *computed* from the observed lepton masses (Eq. (11)). Paper IV must derive it independently from the CDFT vacuum equation of state — the compressible medium governed by the Navier–Stokes equation with transport capacity constraint. When this is done, ϕ_c and hence θ and hence all lepton masses become zero-parameter predictions.

Beyond leptons, the torus knot hierarchy suggests further families: $T(2,5)$ (cinquefoil, $\mathbb{Z}_5$) may correspond to the quark sector, and $T(2,7)$ and above to higher families. These are not addressed here.

8 Conclusion

Four open problems were listed at the end of Papers I and II. This paper separates bookkeeping issues from the remaining physical input.

Problem 1 was a framing error: κ was already derived in Paper I as $\kappa = \chi^2(\ln 8\chi - \beta + 1)$.

Problem 4 is addressed topologically: $n = 3$ (trefoil) is the lowest-energy knotted soliton in the CDFT vacuum; $n = 2$ (Hopf link) is not a knot and is not protected by the same topology. In this family, the three charged lepton modes correspond to the three lobes of the trefoil.

Problem 3 is expressed structurally: $M = \rho_0 c^2 a^3 g(\chi, \beta)$, with $g = 1{,}575.83$ fully determined and ρ_0 the sole unknown.

Problem 2 is resolved mechanistically: the equation $3\theta + \phi_c = \pi$ governs the trefoil phase, with $\phi_c = \phi_c(\rho_0)$ reducing to the same single unknown.

One free quantity remains in the entire lepton sector: ρ_0. Paper IV closes it.

Acknowledgments

The author used AI-assisted tools for formatting checks and code-verification support. The scientific claims, interpretations, and final manuscript decisions are the author's responsibility.

The author is indebted to the foundational work of L. Faddeev and A.J. Niemi [5] (knots and particles), T.H.R. Skyrme [3] (topological solitons), A.F. Rañada [4] (knotted field solutions), G.E. Volovik [7] (vacuum as condensate), and D. Rolfsen [8] (torus knot classification) — whose work provides the mathematical and physical foundations for the topological arguments in this paper.

A Supplementary Material

Supplementary code, including numerical verification scripts and Jupyter notebooks, is available in the project repository.

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